# RateController User Manual

RateController (RC) is a Windows application for controlling the application rate of agricultural products (liquids, granules, seeds, ammonia) in conjunction with AgOpenGPS. It communicates with rate control modules via UDP/Ethernet to manage product application rates, boom sections, and tank quantities.

RC can also be used without AgOpenGPS. Automatic rate control based on ground speed will continue to work normally. Section control will be manual only.

---

## Table of Contents

↑ Contents

## 1. Getting Started

Before operating, complete these setup steps in order:

1. **Create or load a Profile** — a profile stores all machine settings (products, sections, relays, etc.)
2. **Connect the module** via Ethernet — go to **Modules > Communication**, set the subnet, and send it to the module
3. **Load a board preset** — go to **Modules > Boards** and select the Example Preset matching your hardware
4. **Configure the module** — go to **Modules > Config** to set the Module ID, description, and relay type
5. **Configure pins** — go to **Modules > Pins** to verify or set sensor and PWM pin assignments
6. **Configure relay pins** — go to **Modules > Relay Pins** to assign physical pins to each relay
7. **Set valve mode** — go to **Modules > Valves** to select 2-wire or 3-wire valve control
8. **Define sections** — go to **Machine > Sections** and set section widths
9. **Configure relays** — go to **Machine > Relays** and assign relay types to sections
10. **Configure switches** — go to **Machine > Switches** to set master switch behavior
11. **Set up your product** — go to **Products > Rate** and enter product name, control type, units, and rates
12. **Assign the sensor** — go to **Products > Settings** to assign the module ID and sensor ID
13. **Calibrate** — go to **Machine > Calibrate** to determine the Cal Factor
14. **Create a Job** — go to **File > Jobs** to create a job for the field
15. **Connect AgOpenGPS** — the AOG status indicator should turn green

### Data Folder (File > Folders)

RC stores its data — profiles, jobs, fields, rate maps, and logs — in a data folder (Documents by default). Use **File > Folders** to move it, e.g. to a synced folder (OneDrive, Dropbox) or another drive.

| Control | Description |
|---|---|
| **Current location** | The data folder currently in use |
| **New location** | Press to browse for the new data folder; the chosen path can also be typed in the box |
| **Copy Data** | Copy the contents of the current data folder to the new location. Leave unchecked to simply switch to a folder that already contains RC data. |
| **Overwrite** | Allow copying into a folder that already contains files. Without this, Copy Data only copies into an empty folder. |

Press **OK** and confirm — RC changes the folder and restarts. The new location cannot be a sub-folder of the current data folder.

---

↑ Contents

## 2. Profiles

**Menu: File > Profiles**

A profile stores all machine configuration settings. Multiple profiles can be maintained for different implements.

| Button | Action |
| --- | --- |
| 1. New | Create a new profile |
| 2. Load | Load the selected profile |
| 3. Delete | Delete the selected profile. Default, and the last remaining editable profile, can not be deleted |
| 4. Copy | Copy the selected profile to a new name. The copy is always editable, even when copied from Default |
| 5. File name | Enter name for New or Copy |
| 6. Files | List of available profiles to select |
| ↑ ↓ 7. Page Up/Down | Scroll through the profile list |

### The Default Profile

Default is a read only reference profile. It holds a known good starting configuration and is never changed by the program, so it is always available to start over from. It can not be deleted.

Because it is read only, settings changed while Default is loaded are **not saved**. To warn you of this, the profile name at the top left of the menu is shown on a **red background** whenever a read only profile is loaded.

To start a new setup from the reference configuration, select **Default**, type a name, and press **Copy**. The copy is a normal editable profile and is loaded automatically.

> Because Default can not be saved to, the program always keeps one editable profile. On first run it makes one from Default and names it **Example**. Example is an ordinary profile — edit it, copy it, or delete it once you have a profile of your own. It is only rebuilt if no editable profile is left.

---

↑ Contents

## 3. Jobs

**Menu: File > Jobs**

Jobs track application activity per field. Each job stores application data, notes, and a rate map.

### Current Job

| Field | Description |
|---|---|
| Name | Job name |
| Date | Date/time of the job |
| Field | Select from list or type a new field name |
| Notes | Free-text notes for the job |

### Jobs List

The list shows each job's **Name**, **Date**, and **File**. Click a job to select it.

| Control | Action |
|---|---|
| Year | Year used by the filter |
| Field | Field used by the filter |
| Filter | When checked, only jobs matching the selected Year and Field are listed |
| Field | Delete the selected field from the Fields list (not allowed while the field is in use) |
| Load | Open the selected job |
| New | Create a new job |
| Copy | Copy selected job and its rate map to a new file |
| Delete | Delete selected job |
| Show at Start-Up | Display the Jobs menu when the app launches |
| Activity | Job report |
| Import | Import jobs |
| Export | Export all jobs to a chosen folder |

↑ Contents

# 4. Options – Speed Source & Display

**Menu: File > Options**

Selects the source of ground speed used for rate calculations and configures display units.

## Speed Source

| Option | Description |
| --- | --- |
| **GPS** | Use speed from AgOpenGPS |
| **Simulated Speed** | Simulate a constant speed (useful for testing) |
| **Wheel Sensor** | Use a speed sensor connected to a module GPIO pin |

## Wheel Sensor Setup

1. Enter the **Module ID** and **GPIO pin** used for speed sensing.
2. Enter the calibration number in **pulses per mile** or **pulses per kilometer**.
3. Press the **up arrow** to send configuration to the module (also sent on Save).
4. Use the **Calibration screen** to calculate the calibration number if unknown.

## Display Options

| Option | Description |
| --- | --- |
| **Metric** | Toggle between metric (km, ha, L) and imperial (mi, ac, gal) units |
| **Rate Display** | Show or hide an additional rate display window with a larger font |

---

↑ Contents

## 5. Products – Rate

**Menu: Products > Rate**

Configures the fundamental properties of a product.

| Setting | Description |
|---|---|
| **1. Product Name** | Name of the product |
| **2. Control Type** | How the application rate is controlled (see below) |
| **3. Quantity Units** | Units used to measure the product (L, gal, kg, lbs, seeds, etc.) |
| **4. Coverage Units** | Acre, Hectare, Minute, or Hour |
| **5. Sensor Counts/Unit** | Number of sensor pulses per unit of product applied |
| **6. Density** | Product density (used for mass/volume conversion) |
| **7. Base Rate** | Target application rate |
| **8. Alt. Rate (%)** | Alternate rate as a percentage of the base rate |
| **9. Tank Size** | Total tank capacity |

### Control Types

Selects the actuator driving the product and how the PID loop positions it. This must match the physical valve/motor wiring on the module — an incorrect selection can leave the actuator open or unresponsive.

| Type | Description |
|---|---|
| **Standard Valve** | A proportional flow-control valve driven continuously by the PID loop — includes a motorized ball valve or a solenoid valve. PWM = 0 stops the valve motor and holds its last position (does not drive it closed) — use this for valves with no dedicated closed/shutoff position. |
| **Combo Close** | Same continuous PID control as Standard Valve — includes a motorized ball valve or a solenoid valve — but the valve is also driven fully closed whenever the product/section is off (rather than just holding position). Use for valves that must shut off flow, not just stop adjusting. |
| **Motor** | Drives a variable-speed motor that IS the metering device itself (e.g. a hydraulic or electric drive motor turning a metering roller/auger/pump) — not a motorized valve. PID output accumulates over time (integrating control) instead of positioning directly, so it needs different gain tuning than a valve. |
| **Combo Timed** | For close-only/on-off style valves that can't be positioned proportionally. Instead of continuous PWM, the valve is pulsed through alternating Adjust/Pause cycles (set on the Control page) to work the rate toward target, with a fast minimum-start ramp from zero. |

---

↑ Contents

# 6. Products – Settings

**Menu: Products > Settings**

Configures sensor assignment, alarm thresholds, and display preferences per product.

| Setting | Description |
| --- | --- |
| **1. Enable product** | Activate this product |
| **2. Rate Sensor Location** | Module ID (0–7) and Sensor ID (0–1) for the flow sensor |
| **3. Minimum UPM** | Minimum flow rate the controller will maintain. Enter either a fixed **UPM** value or a **By Speed** value (MPH/KPH) — the hint below the boxes shows the equivalent value the other way (e.g. the speed that corresponds to the entered UPM). |
| **4. Off-Rate Alarm** | % error in rate that triggers an alarm |
| **5. Default Product** | Product displayed at startup |
| **6. Scale Weight** | Show the scale weight screen |

## Controls

| Control | Action |
| --- | --- |
| Navigate left | Previous product |
| Navigate right | Next product |
| Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

## 7. Products – Mode

**Menu: Products > Mode**

Sets the operating mode for each product. Navigate between products with the left/right arrows.

| Mode | Description |
|---|---|
| **Controlled UPM** | PID feedback control — adjusts motor/valve to maintain the target rate as speed changes |
| **Constant UPM** | Maintains a fixed flow rate based on total machine width regardless of which sections are on. Used for sprayers where turned-off sections return flow to the tank (after the flow meter). RC corrects the measured flow by subtracting the estimated contribution of the off sections to give an accurate application rate per acre/hectare. |
| **Document Applied** | Tracks and records material that has already been applied — no active control |
| **Document Target** | Tracks and records the target rate for comparison — no active control |
| **Document Applied Manual Rate** | For implements with a fixed mechanical/manual metering that has no flow sensor or module at all (e.g. a fertilizer spreader gate set to a position). Enter the implement's flow output in units per minute in the Manual Rate box; RC computes the displayed/logged application rate from that fixed value and ground speed — no active control, no module. |

**Manual Rate** (enabled only when Document Applied Manual Rate is selected): enter the implement's measured output in units per minute, **not** per acre/hectare — typically found by running the implement at its set position and weighing or measuring output over a timed interval. Because the value is a constant flow, the displayed/logged rate per acre/hectare will rise and fall with ground speed even though the implement setting hasn't changed — this is expected for fixed-metering implements.

### Controls

| Control | Action |
|---|---|
| Navigate left | Previous product |
| Navigate right | Next product |
| Cancel | Discard changes |
| Save | Save settings |

↑ Contents

# 8. Products – Data Reset

**Menu: Products > Data**

Displays accumulated totals for the selected product and allows resetting individual counters. Navigate between products with the left/right arrows.

| Counter | Description |
|---|---|
| **Coverage Area 1** | Total area covered (primary counter) |
| **Coverage Area 2** | Total area covered (secondary counter) |
| **Quantity Applied 1** | Total product applied (primary counter) |
| **Quantity Applied 2** | Total product applied (secondary counter) |
| **Hours 1** | Operating hours (primary counter) |
| **Hours 2** | Operating hours (secondary counter) |

Press the reset button next to any counter to zero it. Both counters are independent, allowing a session total and a season total to be tracked simultaneously.

## Controls

| Control | Action |
|---|---|
| Navigate left | Previous product |
| Navigate right | Next product |
| Reset counter | Zero the selected counter |
| Cancel | Discard changes |
| Save | Save settings |

↑ Contents

# 9. Products – Monitoring

**Menu: Products > Monitoring**

Displays real-time diagnostic data for the selected product. Navigate between products with the left/right arrows.

### Rate Information

| Display | Description |
|---|---|
| **Target UPM** | The current setpoint (desired flow rate) |
| **Applied UPM** | The measured actual flow rate |
| **Rate Error %** | Difference between target and actual, with sign |
| **PWM** | Current PWM output value (0–255) |
| **Frequency (Hz)** | Flow sensor pulse frequency |

### Speed & Coverage Information

| Display | Description |
|---|---|
| **Ground Speed** | Current speed in MPH or KPH |
| **Working Width** | Implement working width in feet or meters |
| **Work Rate** | Area covered per hour (ac/hr or ha/hr) |

### Section Status

Shows 16 section indicator boxes — green for On, red for Off.

### Module Communication

| Display | Description |
|---|---|
| **WiFi / Signal** | Module signal strength (RSSI) bar |
| **Counts per Revolution** | The number of pulses the sensor produces per revolution. Total pulses per minute are divided by this value to calculate RPM. For example, a value of 3 means every 3 pulses equals one revolution. |

### Controls

| Control | Action |
|---|---|
| Navigate left | Previous product |
| Navigate right | Next product |
| Cancel | Discard changes |
| Save | Save settings |

## 10. Products – Control (PID Tuning)

**Menu: Products > Control**

Configures the PID controller that adjusts the motor or valve to maintain the target rate.

### Primary Settings

| Setting | Description |
|---|---|
| **1. Gain** | Rate adjustment factor — how strongly the controller responds to rate errors. The effect is linear: doubling Gain doubles the correction. Errors are measured as a percentage of the target rate, so the same Gain setting behaves consistently at low and high rates. |
| **2. Integral** | Trims residual error once the rate is close to target (inside the BrakePoint band — see below). Far from target, only Gain (proportional) drives the approach; Integral does not speed that up. Increase if the rate settles near, but not exactly on, target. Integral accumulates once per PID loop tick, so a shorter PID loop time (setting 4 below) also makes it trim faster in real time, independent of this setting. |
| **3. Max Power** | Maximum PWM power delivered to motor or valve |
| **4. Min Power** | Minimum PWM power delivered to motor or valve |

### Adjustment Process

1. **Initial Setup** — Set minimum power to the minimum needed to move the motor/valve. Set maximum power to 100. Set Integral to 0.
2. **Gain Adjustment** — Slowly increase gain until the system overshoots the target, then reduce slightly to stabilize.
3. **Integral** — If the rate approaches target but settles slightly off, increase Integral to close the remaining gap. It only acts once you're already close (see BrakePoint) — it will not speed up a slow ramp toward target.

## Optional / Advanced Settings

Settings 7–9 apply to the Combo Timed control type only; setting 5 applies to Motor only.

| Setting | Description |
|---|---|
| **1. Deadband** | Error % below which no adjustment is made. Also controls when accumulated Integral is reset — only once the error clearly crosses to the other side of target, not on small wobble near the setpoint. |
| **2. BrakePoint** | Error % (of peak target) defining the "near target" band. Outside this band, only fast proportional (Gain) control acts and Integral stays at zero. Inside it, proportional gain drops to the Slow Adjustment % and Integral begins accumulating. |
| **3. Slow Adjustment** | % of full proportional adjustment used once inside the BrakePoint band |
| **4. PID loop time** | How often (50–250 ms) the PID recalculates its output. A shorter time makes the valve/motor react sooner to a real rate change, and also speeds up Integral trimming in real time (see Integral, above) since it accumulates once per tick. Reducing this after Gain/Integral were tuned at a slower rate can introduce new oscillation — check for overshoot as with any Gain increase. |
| **5. Slew Rate** | Motor control type only — maximum total PWM change per PID loop. Has no effect on Standard/Combo Close valves, which are positioned fresh each loop rather than ramped. |
| **6. Max Integral change** | Maximum Integral PWM change per PID loop. The total accumulated Integral is separately capped to the Max Power − Min Power range, so it can reach full authority (e.g. to free a stuck valve) without jumping there in one loop. |
| **7. Min Start %** | Combo Timed control type only — % of target rate below which the adjust/pause cycle is skipped for a faster start from 0. |
| **8. Adjust time** | Combo Timed control type only — milliseconds the valve is actively adjusted during each cycle. |
| **9. Pause time** | Combo Timed control type only — milliseconds the valve pauses (holds) between adjustments. |
| **10. Min flow sensor Hz** | Pulse frequency below which the sensor is treated as stopped/no-flow |
| **11. Max flow sensor Hz** | Pulse frequency above which pulses are rejected as sensor noise/bounce |
| **12. Flow sample time (Smp Tm)** | Time window in milliseconds (50–1500) used to calculate the median flow reading. A longer window smooths the reading; a shorter window responds faster. Default 400 ms. |

## Controls

| Control | Action |
| --- | --- |
| Navigate left | Previous product |
| Navigate right | Next product |
| Load defaults | Restore default PID values |
| Open rate graph | Display the rate graph |
| Gain/Integral/Power + | Increase the selected value |
| Gain/Integral/Power - | Decrease the selected value |
| Show advanced settings | Expand the advanced/optional settings panel |
| Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

## 11. Machine – Calibrate

**Menu: Machine > Calibrate**

Use this screen to determine the **Cal Factor** (pulses per unit) for a product. The goal is to achieve a consistent meter roller RPM or liquid flow rate matching typical field operation.

### Step-by-Step Calibration

#### ① Step 1 — Set Ground Speed

- Enter the ground speed you typically operate at.

#### ② Step 2 — Set Meter Roller RPM / Valve Position

- Switch the product **ON** using the power button.
- Enter an **Initial Base Rate** and an **Estimated Cal Factor**.
- Press **Start** — RC adjusts flow to match the target rate.
- If successful, RPM is locked.
- The PWM value used is displayed under the lock (for motor controllers).
- RC stops automatically when the target rate is met.

*If RC fails to reach target RPM:*

- Press **Stop**.
- Check ground speed.
- Adjust the estimated Cal Factor, PWM min/max, or metering drive range.

#### ③ Step 3 — Set Actual Cal Factor

- Press **Start** to run the meter roller and collect a sample.
- Press **Stop**, then enter the measured amount collected.
- RC calculates and displays the new Cal Factor.
- Press **Save** to record it.

#### ④ Step 4 — Optional Adjustments

- To redo meter speed: press **Unlock**, then repeat Step 2.
- To reuse a previously saved setting: press **Lock**, then proceed to Step 3.

### Important Notes

- AgOpenGPS is **not** used during calibration.
- The Switchbox is **not** used during calibration.
- If a switchbox is present, ensure the **master switch is OFF** before calibrating.
- Calibration uses **Constant UPM, Mode 2**.
- All sections on the calibrating module are activated during calibration; modules that are not calibrating are unaffected.

- More than one product can be calibrated at the same time, on the same or different modules (requires up-to-date module firmware).
- After calibration, the original mode and section states are restored.

## Controls

| Control | Action |
| --- | --- |
| Start | Start the meter roller / begin collecting a sample |
| Stop | Stop the meter roller |
| Lock/Unlock | Lock or unlock the current RPM setting |
| Power on/off | Switch the product on or off |
| Cancel | Discard changes |
| Save | Save the calculated Cal Factor |

---

↑ Contents

## 12. Machine – Sections

**Menu: Machine > Sections**

Defines the boom sections and their widths.

- There are **16 sections per module** and up to **8 modules**.
- Module 0 = sections 1–16
- Module 1 = sections 17–32, etc.
- Sections can be defined **individually** or grouped into **zones**.
- All sections in a zone share the same width.
- All sections in a zone are controlled by one switch.
- AgOpenGPS can still auto-control each section individually within a zone.
- When creating a new profile, sections must be defined.
- Sections are also automatically redefined when AgOpenGPS section count changes.

| Control | Action |
|---|---|
| ⊕ Copy Button | Copies the width from Section 1 to all other sections |
| ⊗ Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

# 13. Machine – Relays

**Menu: Machine > Relays**

Assigns relay functions to each relay output on the module. There are **16 relays per module** and up to **8 modules**.

| Control | Action |
|---|---|
| **Reset (circle)** | Reset all relays to default |
| **Renumber (+)** | Renumber sections sequentially |
| **Relay Number** | The relay output number |
| **Relay Type** | The behavior of the relay (see below) |
| **Number** | For Switch type: switch number. For Section/Invert_Section type: section number. |
| Cancel | Discard changes |
| Save | Save settings |

### Application Control

Spraying occurs when:

- Master is ON
- A section is enabled
- In Auto, the guidance system allows application

### Relay Types

| Type | Description |
|---|---|
| **Section** | Controls an individual section.<br>In Manual, the switch turns the section on or off.<br>In Auto, the guidance system controls the section, but the switch can disable it. |
| **FlowMaster** | Controls overall product flow.<br>In Manual, it follows the Master switch.<br>In Auto, it is on only when the Master switch is on and the machine is moving.<br>Use this to control overall product flow through an upstream valve or common shutoff. |
| **Invert_FlowMaster** | Opposite of the FlowMaster output — on when the master switch is off or (in Auto) the machine is stopped. Intended for valve wiring that requires an inverted signal, not as a bypass for section activity. Use **Bypass** if you need a relay that responds to whether sections are actually on. |
| **Bypass** | On when all sections are off; off when any section is on. Use this for a return or bypass valve that diverts product when no boom sections are applying. |
| **Master** | Follows the Master switch.<br>Use this when a relay should follow the master logic itself rather than actual flow-enabled status. |
| **Invert_Master** | Opposite of the Master output (on when the Master output is off). |
| **Slave** | On when any section is on; off when all sections are off. |
| **Invert_Section** | Opposite of the Section output (on when the section is off). |
| **Power** | Always on. |
| **Switch** | Controlled directly by a switch (not affected by Master or Auto). |
| **Hyd Up** | Hydraulic raise. |
| **Hyd Down** | Hydraulic lower. |
| **Tram Right** | Tram marker right. |
| **Tram Left** | Tram marker left. |
| **Geo Stop** | Geofence stop output. |
| **None** | No function assigned to the relay. |

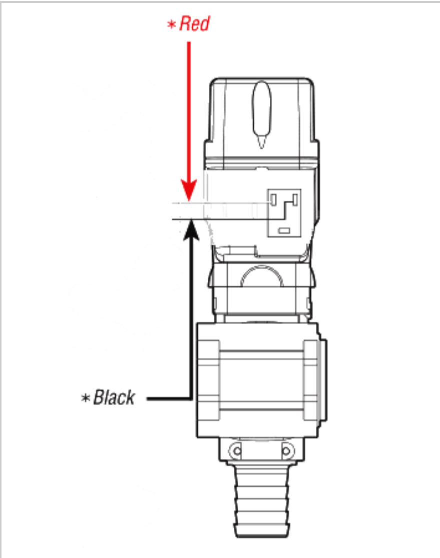
### FlowMaster Valve Options

These options apply only when exactly one **FlowMaster** relay is defined on the selected module.

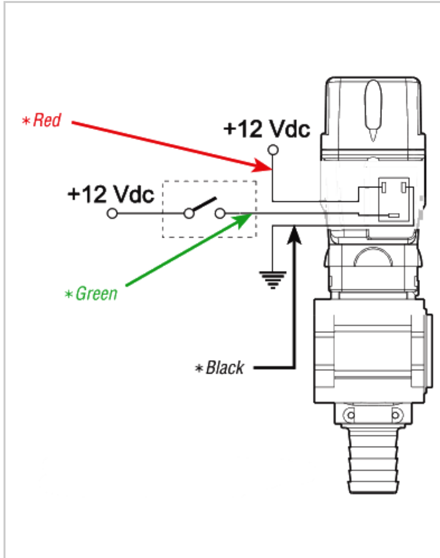
| Option | Description |
|---|---|
| **2-wire valve (single relay powers open and close)** | One **FlowMaster** relay drives both open and close directions using the module's H-bridge output. Requires module support (e.g. RC15). |
| **2-wire valve (2 relays: FlowMaster powers open, Invert_FlowMaster powers close)** | Two relay outputs: the **FlowMaster** relay supplies power to open the valve; the **Invert_FlowMaster** relay supplies power to close it. Use this on modules without H-bridge support (e.g. RC11-2). |
| **3-wire valve (relay signal, valve self-powered)** | One **FlowMaster** relay provides a signal. The valve uses its own power supply to open or close based on that signal. |

### Valve Wiring Reference

The diagrams below show the wiring difference between 2-wire and 3-wire valves.




**2-wire valve**




**3-wire valve**

---

[↑ Contents](#)

# 14. Machine – Switches

**Menu: Machine > Switches**

Configures how the master switch, auto switches, and on-screen virtual switches behave.

### Master Switch Mode

Select one option. Controls what the master switch turns on and off.

| Option | Description |
| --- | --- |
| **Standard** | Master switch controls the master relay and all section relays. When master turns off, all section relays turn off. When master turns back on, section relays resume their switch positions. |
| **Master Override** | Forces the module to treat master as always on, regardless of switch state. Useful for motor control when the master switch should not interrupt the motor PID loop. |

### Switch Type

Select one option. Describes the physical type of master switch connected.

| Option | Description |
| --- | --- |
| **Momentary** | Standard push-to-make switch. Each press toggles master on or off. Holding in the On position activates Primed Start after the configured delay. |
| **Maintained** | Toggle or latching switch. Master follows the switch position directly — on when switch is on, off when switch is off. Primed Start is not available in this mode. |

### Other Settings

| Setting | Description |
| --- | --- |
| **Work Switch** | Uses a switch connected to the implement to gate master on/off. The switch's pin, momentary behavior and NO/NC invert are set per module under Modules > Pins. |
| **Auto Switch – Rate** | Enables automatic rate control via the switchbox Auto Rate switch |
| **Auto Switch – Sections** | Enables automatic section control via the switchbox Auto Section switch |
| **On-Screen Switches – Enabled** | Shows the virtual on-screen switch panel (see below) |
| **On-Screen Switches – 4 / 8 / 12 / 16** | Number of section switches shown on the on-screen switch panel |

### On-Screen Switch Panel

When enabled, a floating switch panel appears on screen. It stays visible and its buttons track the connected switchbox's live state, but every on-screen button (Master, sections, Auto Rate/Section, Rate Up/Down) becomes unresponsive while a physical switchbox is actively connected — the panel is only functional when no physical switchbox is present. **PRM** is the exception: it always activates Primed Start immediately, even while a physical switchbox is connected.

| Button | Function |
| --- | --- |
| **MST** | Toggles master on/off. Green = on, red = off. |
| **PRM** | Activates Primed Start immediately. Green while priming is active. Disabled when Maintained switch type is selected or when the vehicle is moving. |
| ↑ **/** ↓ | Rate adjustment. Hold to increase or decrease rate or PWM. |
| **1 – 16** | Section switches (4, 8, 12, or 16 buttons, as configured). Green = on, grey = off. |
| **Auto Rate** | Toggles automatic rate control. |
| **Auto Section** | Toggles automatic section control. |

> **Note:** Before pressing PRM, turn on the section switches you want to prime. Only sections that are currently switched on will run during priming. Partial priming (some sections only) is valid.

### Controls

| Control | Action |
| --- | --- |
| ⊗ Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

## 15. Machine – Pressure

**Menu: Machine > Pressure**

Calibrates a pressure sensor using two known reference points, and sets a maximum pressure limit per module.

### Calibration Procedure

1. Select the **Module** the pressure sensor is connected to.
2. Set the system to a **known low pressure**.
3. Enter the sensor **reading** and the actual **pressure** value.
4. Set the system to a **known high pressure**.
5. Enter the sensor **reading** and the actual **pressure** value.
6. Enter the **minimum sensor reading** that corresponds to 0 pressure.
7. Press **Show** to display the live pressure screen.

### Max Pressure

Sets the maximum allowed pressure for the selected module. When the measured pressure exceeds this value:

- The application shows a **High Pressure** alarm.
- The limit is also sent to the module, which reduces flow on its own to bring the pressure back below the limit — this protects the system even if communication with the PC is lost. Requires up-to-date module firmware.

If the pressure limit trips repeatedly in a short time, the module treats it as a fault and keeps reducing flow until the **master switch is turned off**, which resets the protection.

Set Max Pressure to **0** to disable the limit.

### Controls

| Control | Action |
|---|---|
| ⓧ Cancel | Discard changes |
| Save | Save settings |

↑ Contents

# 16. Machine – Primed Start

**Menu: Machine > Primed Start**

Allows boom priming or a simulated product application while the tractor is stopped. The tractor must not be moving for primed start to activate.

| Setting | Description |
| --- | --- |
| **On Time** | How long the application runs (seconds) |
| **Speed** | Simulated ground speed used to calculate application rate during priming |
| **Master Switch Delay** | How long to hold the physical master switch in the On position to trigger primed start (seconds). Applies to momentary switchbox only. |
| **Resume** | After On Time expires, if the tractor is moving, continue with normal operation |

## Activating Primed Start

### Physical Switchbox — Momentary Switch

Hold the master switch in the **On** position for the duration of the **Master Switch Delay**. Primed start activates automatically once the delay expires.

### On-Screen Switch Panel

Press the **PRM** button. Primed start activates immediately — no hold required. The PRM button turns green while priming is active.

Before pressing PRM, turn on the section switches you want to prime. Only sections that are currently switched on will run during priming.

### Cancelling

Press the master switch Off, or press the on-screen **MST** button while master is on. Primed start stops immediately.

> **Note:** Primed Start is not available when **Maintained** switch type is selected in Machine > Switches.

## Controls

| Control | Action |
| --- | --- |
| Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

## 17. Modules – Boards

**Menu: Modules > Boards**

Loads a starting pin layout for a known board type.

### Example Presets

These load a starting pin layout for a known board. After loading, edit the Pins, Relay Pins, and Valves pages to match your own board.

| Preset | Microcontroller |
|---|---|
| **Load Nano (RC12-3)** | Arduino Nano |
| **Load Teensy (RC11-2)** | Teensy 4.1 |
| **Load ESP32 (RC15)** | ESP32 |

### Usage

1. Select the preset that most closely matches your module hardware.
2. Press **Save** to apply the default settings. The Save button highlights when a change is pending.
3. Review and adjust settings on the Config, Pins, Relay Pins, and Valves pages.

> **Note:** Loading a preset overwrites the current module pin and relay settings with defaults. Only do this during initial setup or when switching hardware. The module ID is not changed — a preset always applies to the module selected on the Config page.

> **Note:** The board description and module ID are set on the **Modules > Config** page (see Module Identity, §19).

---

↑ Contents

# 18. Modules – Communication

**Menu: Modules > Communication**

Configures how the application communicates with rate control modules. Select either **Ethernet** or **CAN Bus** as the communication type — the two are mutually exclusive. The settings for each are on the **Ethernet** and **CanBus** tabs.

| Option | Description |
|---|---|
| **Ethernet** | Modules communicate via UDP over the local network. Default mode. |
| **CAN Bus** | Modules communicate via a CAN adapter connected to this PC. Ethernet remains active for AgOpenGPS. |

### Ethernet

Select the PC network interface used to communicate with modules.

| Control | Description |
|---|---|
| **IP Address dropdown** | Lists active network interfaces on this PC. Select the one on the same subnet as the modules. |
| **Module Subnet** | Shows the current subnet the application is using to reach modules. |
| **Send Subnet** | Sends the selected IP subnet to the module so it knows which address to respond to. |
| **Rescan** | Refreshes the IP address list and module subnet display. |

### CAN Adapter

| Driver | Description |
|---|---|
| **SLCAN** | Serial port-based CAN adapter. Requires COM port selection. |
| **InnoMaker USB2CAN** | Native USB CAN adapter (InnoMaker hardware). No COM port required. |
| **PCAN** | Native USB CAN adapter (Peak PCAN hardware). No COM port required. |

### COM Port (SLCAN only)

1. Use the dropdown to select the COM port the SLCAN adapter is connected to.
2. Press **Refresh** to rescan available COM ports.

### Diagnostics

| Option | Description |
|---|---|
| **Diagnostics** | Enables CAN traffic debug logging in the activity log |

### Status Indicators

| Indicator | Meaning |
| --- | --- |
| **Connected** | Green — module data received within the last 2 seconds |
| **Driver Found** | Green — CAN adapter is open and receiving frames within the last 4 seconds |

> **Note:** Changing the CAN adapter type or COM port while CAN Bus is active will restart the CAN bridge with the new settings.

---

↑ Contents

## 19. Modules – Config

**Menu: Modules > Config**

Configures the hardware settings of the connected module.

### Module Identity

The underlined line at the top shows which module the module pages are editing — the module number, the board's description (or reported label), and its connection state. It updates live while the page is open. If it turns red with **"Two boards are answering as module N!"**, two powered boards share the same module ID — disconnect one and give it its own ID before continuing.

| Setting | Description |
|---|---|
| **Module ID** | The module to edit. All module pages (Config, Communication, Pins, Relay Pins, Valves) and the Send button work on this module; other connected modules ignore the settings. |
| **New Module ID** | Leave blank unless changing the board's module ID. Enter the new ID and press Save; the next Send assigns it to the board. Because every board that hears an ID assignment takes the new ID, only one board may be connected — Send refuses otherwise. After the board restarts and reconnects, press Send once more so the description and sensor pins reach it. |
| **New Board Description** | Leave blank to keep the current description. Up to 16 characters (e.g. "Left Tank"), sent to the module with Send and stored in the module's own memory, where it is kept through firmware updates. |

> **Note:** A board with no description reports its built-in hardware (MAC) number instead, e.g. **MAC 04E9E5A1B2C3** — unique to every board. This is how RateController tells new boards apart and detects duplicate module IDs. Give each board a friendly description once it is set up. (Nano boards do not report a label.)

### 1. Relay Type

| Value | Description |
|---|---|
| 0 | No relays |
| 1 | GPIOs — use micro-controller pins directly |
| 2 | PCA9555 — 8-relay module |
| 3 | PCA9555 — 16-relay module |
| 4 | MCP23017 |
| 5 | PCA9685 |
| 6 | PCF8574 |

## 2. Pin Assignments by Board

| Parameter | RC5 (Nano) | RC8 (Nano) | RC11 (Teensy) | RC11-2 (Teensy) | RC12 (Nano) | RC12-3 (Nano) | RC15 (ESP32) |
|---|---|---|---|---|---|---|---|
| Sensor Count | 2 | 2 | 2 | 2 | 1 | 2 | 2 |
| Relay Control | 4 | 4 | 1 | 1 | 2 | 4 | 5 |
| Flow 1 | 2 | 2 | 28 | 28 | D3 | 3 | 17 |
| Flow 2 | 3 | 3 | 29 | 29 | — | 2 | 16 |
| DIR 1 | 4 | 4 | 37 | 37 | D6 | 4 | 32 |
| DIR 2 | 6 | 6 | 14 | 14 | — | 6 | 25 |
| PWM 1 | 5 | 5 | 36 | 36 | D9 | 5 | 33 |
| PWM 2 | 9 | 9 | 15 | 15 | — | 9 | 26 |
| Work Pin | — | — | — | 30 | — | 15 | — |
| Pressure Pin | — | — | — | 40 | — | 14 | — |

## 3. Nano Pin Reference

| Pin Name | Number |
|---|---|
| D0–D13 | 0–13 |
| A0–A7 | 14–21 |

## Controls

| Control | Action |
|---|---|
| ⊗ Cancel | Discard changes |
| Save | Save settings |

↑ Contents

## 20. Modules – Pins

**Menu: Modules > Pins**

Manually assigns the GPIO pins used for sensors and motor/valve control on the module. The grid shows one row per sensor (the sensor count is set in Modules > Config). Pin values range from 0–50. A blank value (displayed as "–") means the pin is not assigned.

| Grid Column | Description |
|---|---|
| **Sensor** | Sensor/product number (read-only) |
| **Flow** | Input pin for the flow sensor |
| **Dir** | Output pin for the motor/valve direction channel |
| **PWM** | PWM output pin for the motor/valve |
| **Bin** | Input pin for the bin-empty sensor. Leave unassigned ("–") if the product has no bin sensor — the bin alarm is off for that product. |
| **Invert** | Inverts the bin sensor reading. Unchecked: an open input reads *empty* (sensor closes to ground while material is present). Checked: a grounded input reads *empty*. |

| Setting | Description |
|---|---|
| **Work Pin** | Input pin for the work/implement switch |
| **Momentary** | Enables momentary switch behavior on the work pin |
| **Invert Work Switch** | Inverts the work switch reading to suit the sensor type. Unchecked: an open input reads as *working* (normally-closed sensor). Checked: a grounded input reads as *working* (normally-open sensor). In momentary mode the toggle occurs on the opposite edge. Older module firmware ignores this setting. |
| **Pressure Pin** | Analog input pin for the pressure sensor |

Press **Rescan** to clear all pin assignments. Duplicate pin numbers are rejected on Save.

> **Tip:** When a board type is selected in Modules > Config, pressing **Defaults** populates these pin values automatically. Only change them if your hardware differs from the standard layout.

### Bin-Empty Alarm

When a Bin pin is assigned, the module watches that input and reports the bin state to RateController. The bin sensor input uses an internal pull-up, so a simple switch or level sensor wired between the pin and ground is all that is required. The module filters the reading — the sensor must hold its state for about 2.5 seconds before the reported state changes, so brief flickers from material sloshing are ignored.

- The alarm is only raised **while the master switch is on** — an empty bin at start-up does not alarm until you begin applying.
- When it triggers, the alarm button shows **Bin Empty** with the product name, the product's tile turns red, and on the selected product the quantity caption changes to **Bin Empty** with the quantity value shown as a red 0.

- The displayed 0 is display-only — the recorded tank amount and quantity applied are not changed, and the real value returns as soon as the alarm clears.
- The alarm clears immediately when the bin refills or the master switch is turned off. Tap the alarm button to silence the buzzer.

> **Tip:** Use a plain digital pin for the bin sensor. Inputs with analog conditioning on the board (such as the pressure input) load the internal pull-up and may not read reliably as a digital input.

### Controls

| Control | Action |
|---|---|
| Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

## 21. Modules – Relay Pins

**Menu: Modules > Relay Pins**

Assigns the physical GPIO pins to each of the 16 relay outputs on the module. Pin values range from 0–50. A value of 255 (displayed as "–") means the relay pin is not assigned.

- **Low Relays (1–8):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.
- **High Relays (9–16):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.

Press **Rescan** to clear all relay pin assignments.

> **Tip:** When a board type is selected and **Defaults** is pressed in Modules > Config, relay pins are populated automatically for the chosen hardware.

### Controls

| Control | Action |
|---|---|
| ⊗ Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

# 22. Modules – Valves

**Menu: Modules > Valves**

Selects the wiring mode for the valve or motor driver.

| Option | Description |
| --- | --- |
| **2-Wire** | Motorized or latching valve — the driver actively powers the valve in both directions to open and close it. |
| **3-Wire** | One switched signal wire, one power wire and one ground wire. Compatible with standard solenoid valves and motorized ball valves that manage open/close internally. Default for most installations. |

Select the mode that matches how your valve or motor driver is wired to the module.

### Controls

| Control | Action |
| --- | --- |
| Cancel | Discard changes |
| Save | Save settings |

---

↑ Contents

## 23. Rate Map

**Menu: File > Rate Map**

The Rate Map enables variable rate application by defining geographic zones with different target rates. It integrates with AgOpenGPS for GPS-referenced mapping.

### Prescriptions Tab

Manages prescription shapefiles for the current field. The active prescription determines which zone rates are used during application.

| Control | Action |
|---|---|
| **Field** | Shows the field name assigned to the current job |
| **Prescription list** | Lists all prescription files available for the current field. Select one to make it active. **Zones.shp** is the default and cannot be deleted. |
| **New** | Create a new empty prescription file with a given name |
| **Copy** | Copy the selected prescription to a new name. The copy becomes the active prescription. |
| **Import** | Open the prescription import dialog — imports a shapefile or an XML prescription. See Importing a Prescription below. |
| **Export** | Export the selected prescription shapefile to a chosen location |
| **Delete** | Delete the selected prescription file. Zones.shp cannot be deleted. |

> **Note:** Selecting a prescription from the list immediately makes it the active prescription for the current job. The map reloads to show the selected zones.

### Importing a Prescription

The Import button opens a two-step import dialog:

#### Step 1 — Import Data

| Control | Description |
|---|---|
| **Shape file / XML** | Choose the source format: a prescription **shapefile** (.shp) or an **XML** prescription file |
| **Import** | Select the file and read it in |
| **Mapping grid** | For shapefiles, match each product (A–E) to the attribute column in the shapefile that holds its rate. For XML files, the layer names found in the file are shown. |
| **Default column** | Fallback rate per product (shapefile import only). Zones missing the mapped attribute receive this rate instead of 0. Tap the cell to enter a value; leave it at 0 for no fallback. |
| **Product** | The primary product (A–E) used when adjusting zones in Step 2. Selected automatically from the imported data; change it if needed. |
| **Name** | File name for the saved prescription |

**Step 2 — Adjust Zones (optional)**

Grid-style prescriptions can contain thousands of tiny zones. Adjust consolidates them into fewer, larger zones that are easier to work with and apply.

| Setting | Description |
|---|---|
| **Max Zones** | Maximum number of distinct rate zones to create |
| **Min zone size** | Zones smaller than this area (Ac/Ha) are merged into their neighbours |
| **Min zone step** | Minimum rate difference between zones — rates closer together than this are combined |
| **Adjust** | Run the consolidation and report how many zones were created |

Press **Save** to save the prescription for the current field. It becomes the active prescription. If the imported data has different product names, RC offers to update the job's product names to match.

## Zones Tab

Create and edit rate zones within the active prescription.

Create zones by clicking the **New** button, then clicking at least **3 points** on the map to define the boundary.

- With **overlapping zones**, the **last zone added** takes priority.
- When variable rate is enabled but **no zone exists**, the **base rate** is used.
- Switching to another tab ends an active **New**/**Edit** — unsaved zone changes are discarded.

| Button | Action |
|---|---|
| New | Create a new rate zone |
| Edit | Edit zone properties (name, rate per product, color) |
| Delete | Delete corner point (while creating) or delete zone (while editing) |
| Cancel | Cancel current edit |
| Save | Save current edit |
| Zone List | Open the Zone List — view all zones with their areas, product rates (A–E), and colors; tap a rate cell to change it; select/deselect zones; delete zones; print the list. To set one rate on many zones at once, check the wanted zones, choose the product (A–E), enter the rate and press **Apply** — then Save or Cancel as usual. |

## Data Tab

| Control | Action |
|---|---|
| Record count | Number of applied rate data points recorded |
| A / B / C / D / E | Select which product's applied rate to display on the map |
| Record | Start/stop recording applied rates |
| Delete | Delete all recorded applied rate data |

### VR Tab — Variable Rate Look-Ahead

Configures how far ahead of a zone boundary the application rate begins adjusting, so the rate matches the target as soon as the machine enters the new zone. Compensates for applicator response lag.

| Setting | Description |
| --- | --- |
| **Look-Ahead Time (per product)** | Seconds before entering a zone boundary that rate adjustment begins. One value per product (up to 5). |
| **Auto Tune** | RC monitors look-ahead performance and automatically adjusts the value to improve zone entry accuracy. |

### Files Tab

Import and manage supporting data files for the current field. Use the file selector dropdowns to choose which file is active for each overlay layer.

#### Field Boundary

The field boundary defines the operated area and is used to clip automatically generated productivity zones. Select the active boundary from the **KML** dropdown. Import a boundary using the **Import Boundary** button — three file types are accepted:

- **AOG Field.kml** — the field KML from the AgOpenGPS Fields folder; the boundary polygon is extracted automatically.
- **AOG Boundary.txt** — AgOpenGPS boundary file; easting/northing coordinates are converted to lat/lon using the field origin.
- **Any KML (*.kml)** — any KML file containing a polygon boundary.

> **Tip:** Field files are typically found in:
>
> - **AgOpenGPS:** Documents\AgOpenGPS\Fields\[Field Name]\ — contains `Field.kml` and `Boundary.txt`
> - **TWOL:** Documents\TWOL\Fields\[Field Name]\ — contains `Boundary.txt`

### Yield, EC, and Elevation Data

| Control | Action |
| --- | --- |
| **Yield dropdown** | Select the active yield data file for the overlay and zone creation |
| **Import Yield** | Import a yield data CSV file. If the file contains elevation data, RC will offer to extract it to a separate elevation file. |
| **Delete Yield** | Remove the selected yield data file |
| **EC dropdown** | Select the active EC (electrical conductivity) data file |
| **Import EC** | Import an EC data CSV file (Lat, Lon, EC) |
| **Delete EC** | Remove the selected EC data file |
| **Elevation dropdown** | Select the active elevation data file |
| **Import Elevation** | Import an elevation data CSV file (Lat, Lon, Elevation) |
| **Restore Elevation** | Restore the previous elevation file from the automatic backup |
| **Delete Elevation** | Remove the current elevation data file |

### Create Zones

The **Create Zones** button opens the productivity zone creation dialog. Zones are generated automatically from yield history, soil EC, elevation data, or any combination. Clear existing zones before generating if you want to replace them.

| Setting | Description |
| --- | --- |
| **Yield files (checklist)** | Check one or more yield data files to include. Multiple years are blended by their assigned weight. |
| **EC** | Include the active EC layer in zone generation |
| **Elevation** | Include the active elevation layer in zone generation |
| **Weights** | Percentage weight for each active layer. All weights must sum to 100 before Build is enabled. |
| **Productivity levels** | Number of productivity classes to create (e.g. 3 = Low / Med / High). Each class may produce more than one zone polygon if the field has non-contiguous areas of the same productivity level. |
| **Min zone size** | Minimum zone area in hectares (or acres). Zone polygons below this size are discarded. Increase to reduce the number of small fragments. |
| **Build** | Generate zones using the selected layers and settings. The field boundary (set in the KML dropdown) clips zones to the operated area. |

> **Tip:** Set the field boundary KML before generating zones. Without a boundary, zones are clipped to the bounding box of the data files, which may not match the actual field edge.

## Display Toggle Buttons (right panel)

These buttons highlight green when active. Click to toggle on or off.

| Button | Action |
| --- | --- |
| Sat. View | Toggle satellite background image |
| Target Zones | Toggle rate zone boundaries display |
| Applied Rates | Toggle applied rate colour overlay |
| KML Files | Toggle KML file overlay |
| Yield Data | Toggle yield data colour overlay |
| EC | Toggle soil EC colour overlay |
| Elev. | Toggle elevation colour overlay |

### Bottom Controls

| Button | Action |
| --- | --- |
| Minimize | Shrink the map to a small window. Drag it by its title bar to reposition. |
| Centre | Centre the map on the current GPS position |
| Help | Open the help page |
| OK | Close the Rate Map |

---

↑ Contents

## 24. Rate Graph

**Menu: Help > Rate Graph** (opens as a separate window)

Displays a real-time chart of the PID control loop for a product. Used for tuning Gain and Integral settings.

### Displayed Data

| Element | Description |
|---|---|
| **Target UPM line** | The desired flow rate setpoint |
| **Applied UPM line** | The measured actual flow rate |
| **Error %** | Current deviation from target |
| **PWM value** | Current PWM output (0–255) |
| **Frequency (Hz)** | Current flow sensor frequency |

### Chart Controls

| Control | Description |
|---|---|
| **Auto** | Auto-scale the Y-axis to fit the data |
| **Up** | Increase Y-axis range by 1.5× |
| **Down** | Decrease Y-axis range by 0.67× |

The chart maintains the last 30 data points. Click the graph to select which product to display.

---

↑ Contents

## 25. Scale / Tank Display

**Menu: Products > Settings > Scale Weight** (enable checkbox to show)

Displays a floating window showing tank/scale data for a product. Each product has its own scale display window.

### Display Modes

Left-click on the displayed value to cycle through four modes:

| Mode | Description |
|---|---|
| **Current Weight** | Net weight from the scale sensor (after tare) |
| **Applied Weight** | Difference between starting weight and current weight |
| **Area Covered** | Area worked since the last reset |
| **Application Rate** | Applied weight divided by area covered |

### Controls

| Action | Result |
|---|---|
| Left-click on value | Resets the starting weight and acres reference point |
| Right-click on value | Resets the tare (zero calibration) |
| Right-click and drag | Repositions the window |

---

↑ Contents

# 26. Language Selection

**Menu: File > Language**

Changes the display language of the application.

### Supported Languages

- English
- Deutsch (German)
- Hungarian
- Nederlands (Dutch)
- Polish
- Russian
- French
- Lithuanian

> **Note:** A restart of the application is required for the language change to take effect.

---

↑ Contents

## 27. Color Customization

**Menu: File > Color**

Customizes the foreground and background colors of the application interface.

### Preset Color Schemes

Three built-in color presets are available. Select one to apply it immediately.

### Custom Color

Select **Custom Color** to open the color picker:

1. Click in the **color gradient panel** to select a hue and saturation.
2. The horizontal position controls **hue**.
3. The vertical position controls **brightness/saturation**.
4. Separate foreground and background colors can be selected.

Color preferences are saved and applied immediately across all screens.

---

↑ Contents

## 28. Help – Logs & Diagnostics

**Menu: Help** (main menu)

Provides access to application logs, module diagnostics, and system information.

### Log Files

| Log | Description |
|---|---|
| **Activity Log** | Records application events and UDP/CAN communication activity in real time |
| **Error Log** | Records application errors and exceptions |
| **Ethernet Log** | Records raw UDP packet traffic — useful for diagnosing communication issues |

Press **Start/Pause** to freeze or resume live log updates.

### Activity Log — Module Status Events

When a module connects or its status changes, RC logs the following to the Activity Log automatically:

| Log Entry | Description |
|---|---|
| Module X, Ethernet connected: True/False | Whether the module's Ethernet port is connected |
| Module X, Wifi Strength: 0–3 | ESP32 WiFi signal level — 0 = not connected, 1 = weak (RSSI < -80), 2 = moderate (RSSI < -70), 3 = good (RSSI < -65) |
| Module X, Valves are 2 Wire / 3 Wire | Valve wiring mode reported by the module |
| Module X, Pin Configuration correct / not correct | Whether the module's pin configuration is valid |

These entries are written once on first connect and again whenever the status changes.

### Controls

| Control | Action |
| --- | --- |
| Start/Pause log | Freeze or resume live log updates |
| Reset logs | Delete all log files after confirmation |
| **PID Logging** | Toggle PID diagnostic logging. While on (green), the module streams per-loop rate control data (target, applied rate, PWM, error) which is saved to a CSV file in the `PIDLogs` folder inside the data folder — useful for tuning and diagnosing rate control problems. The setting is remembered between sessions. |
| Open rate graph | Display the rate graph |
| Check for updates | Check for a newer version of the application |
| Open user manual | Open the user manual |
| Navigate modules left | View previous module diagnostics |
| Navigate modules right | View next module diagnostics |

### Module Diagnostics

| Display | Description |
| --- | --- |
| **Module ID** | ID of the module currently being viewed (use left/right arrows to switch) |
| **Firmware Version** | Firmware version running on the selected module |
| **Elapsed Time** | Seconds since last module data was received |

### System Information

| Display | Description |
| --- | --- |
| **IP / Subnet** | Current Ethernet subnet in use |
| **Application Version** | Installed RC software version and date |
| **Current File** | Name of the active profile file |
| **GPS Coordinates** | Live latitude and longitude from AgOpenGPS |

### External Links

Links to the project repositories are available from this screen:

- AOG RC GitHub repository
- PCB Setup GitHub
- Rate Control PCBs GitHub

## 29. Tips & Tricks

### Main Screen Navigation

- **To reposition the main screen**, press and hold (or right-click) in the middle of the screen, then drag it to the desired location.
- Use the **minimize button** on the main form to reduce the display to rate only. RC remembers this — if it was minimized when closed, it starts minimized next time.

### Clickable Display Elements

- **Click on target rate label** to switch between the target rate and the alternate rate (a percentage of the target). When variable rate is enabled, the label shows the VR target rate and is not clickable.
- **Click on quantity applied label** to switch between quantity applied and quantity remaining.
- **Click on area applied label** to switch between area applied and area remaining.
- **Click on area applied value** to reset the area counter.
- **Click on the tank bar graph** to reset the tank quantity.
- Click a **product icon** at the top of the screen to select which product's rates and totals are displayed. The rate graph will reflect the currently selected product.

---

↑ Contents

## 30. Troubleshooting

### No Relays Activating

1. Is the module Config set correctly for board type and relay type?
2. Has the subnet been set in the app and uploaded to the module?
3. Have section count and widths been set?
4. Is Auto mode on?
5. Is a section assigned to each relay?
6. Is a switch assigned to each section?
7. If Auto is on, have sensor counts/unit and base rate been set?
8. If Work Switch is enabled (Menu > Switches), the master will only activate when the work switch reads on — check the switch's wiring, pin assignment, and the **Invert Work Switch** setting (Modules > Pins) against whether the sensor is normally-open or normally-closed.

### No UPM (No Flow Reading)

1. Check ground speed.
2. Check working width.
3. Check sensor control settings.
4. Is AgOpenGPS connected?
5. Is the module connected?
6. Check sensor counts/unit.
7. Check base rate.
8. Have control values been sent to the module?
9. Check control type — is it a valve or motor?
10. Is the connector fully plugged in?
11. Is the module running the latest firmware?
12. Is Auto mode on?
13. If Work Switch is enabled, is it reading on? A wrong **Invert Work Switch** setting (Modules > Pins) will hold the master off with no flow, even though everything else is configured correctly.

### Document Applied Manual Rate (Mode 5)

1. Don't assign a real module to a product set to Mode 5 — the module's real sensor data will conflict with the manually entered value.
2. Enter the flow value in units per minute, not per acre/hectare; there is no rate alarm in this mode to catch a units mistake.
3. Applied rate will rise and fall with ground speed even though the flow input is constant — this is expected for a fixed-metering implement.
4. A ground speed source is still required — GPS (via AgOpenGPS), a wheel-speed sensor module, or a manually entered/simulated speed (Menu > Speed) all work; with no speed source active, quantity won't accrue even though the flow value is set.

### AOG Status Light Not Green

1. The AOG light reflects whether AgOpenGPS is sending GPS or AutoSteer position data over the network — RateController does not need to send anything for it to turn green.
2. AgOpenGPS only sends that data once it has an active position source: a real GPS fix, or its built-in GPS Simulator turned on. No fix and no simulator means the light stays red, even if RateController and AgOpenGPS are both running correctly.
3. If your AutoSteer hardware also supplies GPS (common on combined GPS/IMU/steer boards), unplugging it removes the position source too — the AOG light will drop even though AutoSteer itself isn't what RateController is waiting on.
4. To test the connection without field hardware, turn on AgOpenGPS's GPS Simulator — the light should turn green the same way it would with a real fix.

### No Rate Adjustment

1. In manual mode, the master switch must be on.
2. Is minimum PWM too low?

### Motor Won't Shut Off

1. Minimum PWM should be set to 0.
2. Is master override enabled?

### After Updating Module Firmware

1. Resend the subnet (Modules > Communication).
2. Resend all module config settings (Modules > Config, then Send).

↑ Contents